

Carlos Fernando Revilla *Software Engineer | M.S. Data Science*

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TECHNICAL SKILLS

Programming Languages: Python, C# (.NET), JavaScript, TypeScript, Java, C++, C, SQL

Software Tools: AWS, Terraform, Docker, CI/CD, Quartz.NET, .NET MVC, Apache Kafka & Spark, PostgreSQL, MongoDB, Solr, ETL/ELT Pipelines, TensorFlow, LLM Integration, Git, Jest

WORK EXPERIENCE

Philips | Full-Stack Software Engineer

Cambridge, MA | January 2026 - Present

- Enabled on-demand replication of a 20-service AWS architecture across regions and environments by designing a Terraform-based per-tenant isolation pattern for 5 services, then migrating and scaling it to AWS CloudFormation.
- Automated multi-service deployments with GitHub Actions and Docker, using OIDC-based AWS authentication to support zero-downtime releases.
- Delivered resilient, dynamically-scheduled data aggregation for patient monitoring utilization data, with automatic recovery from failed jobs via built-in retry logic, by building a multi-tenant background processing service in C#/.NET 8 with Quartz.NET cron-based scheduling synchronized from PostgreSQL, aggregating data across AWS RDS and Citus.
- Enhanced an internal React/Vite dashboard for monitoring real-time product data streams, using AI-assisted development tools to accelerate feature delivery.

Silverchair | Full-Stack Software Engineer

Charlottesville, VA | May 2024 - February 2025

- Delivered interactive learning features including quizzes, study modules, and interactive diagrams for a client-facing medical-education platform, by developing new functionality within a .NET MVC architecture serving textbook content to medical students.
- Reduced web and mobile search latency by ~300ms and cut database load for textbook search and filtering by title, quiz, author, and ISBN, by eliminating N+1 query bottlenecks in LINQ keeping queries as IQueryable until final projection and tuning mobile middleware functions with caching and batching.
- Improved content accuracy and reduced user-reported errors by 28% by diagnosing data integrity issues in an Apache Solr pipeline and re-ingesting missing columns.
- Reduced manual QA efforts by 2 hours per week by building a comprehensive Jest test suite for data-critical features, strengthening JavaScript application reliability.

EDUCATION

M.S. - Data Science | University of Virginia

- *Focus:* Machine learning, Data engineering, Big Data Systems, and Natural Language Processing.

B.A. - Computer Science | University of Virginia

- *Minor: Data Science - Relevant Courses:* Artificial Intelligence, Data Science Systems, Database Systems

Software Engineering Bootcamp | Codesmith, Full-time Immersive

- *Focus:* Advanced JavaScript, TypeScript, Database Design, State Management, React + Redux Toolkit

PROJECTS

OS-Labs: KafkaNightOwl | Full-Stack Software Engineer

- Streamlined Kafka message transactions by building a Message Data & Metrics Service with Node.js/Express.
- Built a maintainable web dashboard for Kafka message metrics by developing a React/TypeScript interface using Flux architecture and container patterns.

Acne-AI | Full-Stack Software Engineer

- Achieved strong benchmark performance on a held-out test set by fine-tuning Meta's DINOv2 foundation model for skin condition classification using Python and Google Colab A100 GPUs.
- Reduced diagnosis latency by integrating a React frontend with a Python model backend using Axios and optimizing image processing.

TALKS & PUBLICATIONS

GraphQL & the N+1 Problem | Speaker

- Presented solutions to GraphQL's N+1 query issue. Talk sponsored by Codesmith, Jeeny, and Bractlet.